

DESIGN AND IMPLEMENTATION OF AN EDUCATIONAL ELECTRICITY LOAD FORECASTING SYSTEM: A MODULAR AND ADAPTIVE APPROACH

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INTRODUCTION

This project develops a lightweight, educational electricity load forecasting app. Using a Random Forest model and basic time-based features (hour, day, year), it allows users to import data, train the model, generate predictions, and visualize results via a simple GUI built with tkinter. The system emphasizes accessibility and learning, demonstrating the complete machine learning workflow—data handling, feature extraction, modeling, and forecasting—without requiring advanced programming or large-scale infrastructure.

GOAL

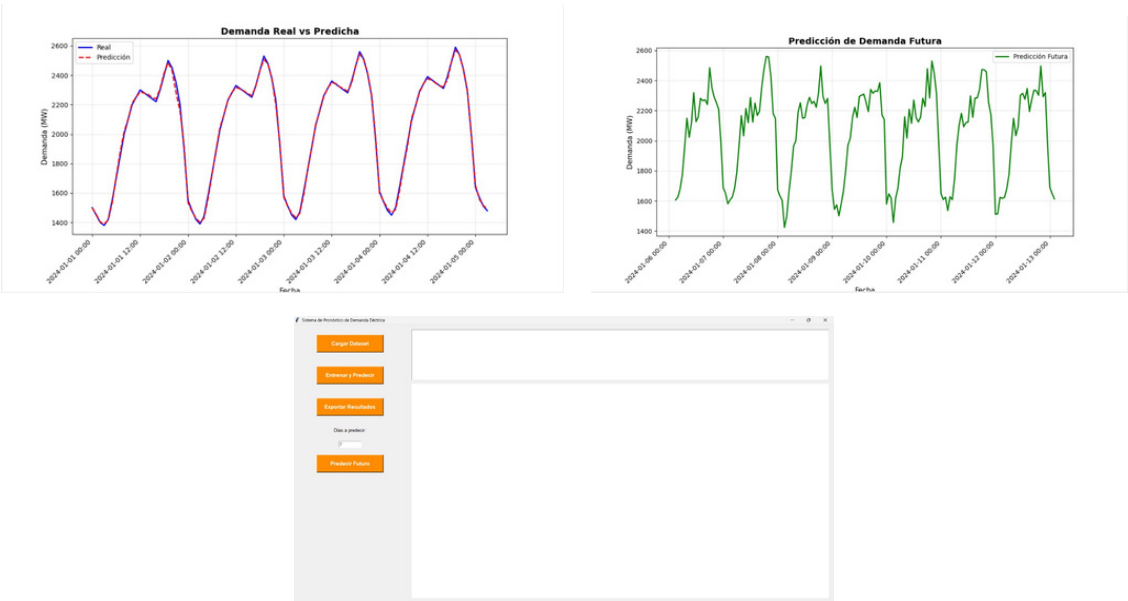
Develop an educational electricity load forecasting app that demonstrates the full machine learning workflow—data handling, feature extraction, modeling, and prediction—through an intuitive GUI, allowing users to learn and experiment without advanced programming skills.

METHODS

This project implements a complete end-to-end electrical load forecasting system using a Model–View–Controller (MVC) architecture. The Model handles data loading, preprocessing, temporal feature extraction, and training a Random Forest regressor. The View provides a user-friendly tkinter GUI with buttons, input fields, progress bars, message logs, and embedded plots, allowing interaction without programming knowledge. The Controller manages the workflow, coordinating training, historical predictions, and multi-step future forecasts. The system automatically handles missing value imputation, datetime conversion, temporal feature creation, and temperature normalization. Users can load CSV datasets, visualize historical and predicted loads, forecast future demand using synthetic temperature data, and export results as CSV files. This structure ensures modularity, maintainability, and an accessible workflow, demonstrating the full machine learning pipeline in a practical and interactive way.

RESULTS

The forecasting application was evaluated to verify correct behavior of all modules: data loading, preprocessing, Random Forest model, and GUI. Unit tests confirmed that each component worked independently, including CSV import, missing value handling, temporal feature generation, model training, prediction, and file export, with all tests passing successfully. Integration tests through the GUI showed consistent end-to-end behavior for datasets of different sizes: files were correctly loaded, preprocessing and feature generation ran without errors, the model trained efficiently, predictions were displayed in the interface, and outputs could be exported smoothly. Visual inspection of time-series plots confirmed that the Random Forest model captured main temporal patterns, producing coherent forecasts aligned with the dataset’s trends. Overall, the system proved stable, reliable, and functional across all workflows.



FUTURE DIRECTIONS

- Future improvements could include:
- Using additional ML models (XGBoost, LSTM, etc.)
 - Adding evaluation metrics (RMSE, MAE, MAPE)
 - Including more explanatory variables (humidity, holidays, etc.)
 - Implementing model comparison and hyperparameter optimization
 - Enhancing visualizations
 - Extending to web or cloud platforms

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